MSG No. 59.412 P_bmmn [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `.2'/m.`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ 0\right]$	`[1,7,11,13]`
2	$\left[\frac{1}{2},\ \frac{3}{4},\ 0\right]$	`[2,8,12,14]`
3	$\left[0,\ \frac{3}{4},\ 0\right]$	`[3,5,9,15]`
4	$\left[\frac{1}{2},\ \frac{1}{4},\ 0\right]$	`[4,6,10,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,5,11,15]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,6,12,16]`
3	$[0,\ 0,\ 0]$	`[3,7,9,13]`
4	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[4,8,10,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `.2'/m.`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ \frac{1}{2}\right]$	`[1,7,11,13]`
2	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[2,8,12,14]`
3	$\left[0,\ \frac{3}{4},\ \frac{1}{2}\right]$	`[3,5,9,15]`
4	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[4,6,10,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,5,11,15]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,6,12,16]`
3	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3,7,9,13]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,8,10,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[2,3,5,8]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[9,12,14,15]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[10,11,13,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `mm'2'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	`[1,6,12,15]`
2	$\left[\frac{3}{4},\ \frac{1}{2},\ -z\right]$	`[2,5,11,16]`
3	$\left[\frac{3}{4},\ 0,\ -z\right]$	`[3,8,10,13]`
4	$\left[\frac{1}{4},\ 0,\ z\right]$	`[4,7,9,14]`

Table 7: Wyckoff site: `8g`, site symmetry: `.2'.`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,11]`
2	$\left[\frac{1}{2},\ -y,\ 0\right]$	`[2,12]`
3	$\left[0,\ y+\frac{1}{2},\ 0\right]$	`[3,9]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[4,10]`
5	$[0,\ -y,\ 0]$	`[5,15]`
6	$\left[\frac{1}{2},\ y,\ 0\right]$	`[6,16]`
7	$\left[0,\ \frac{1}{2}-y,\ 0\right]$	`[7,13]`
8	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[8,14]`

Table 8: Wyckoff site: `8h`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,11]`
2	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	`[2,12]`
3	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[3,9]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[4,10]`
5	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[5,15]`
6	$\left[\frac{1}{2},\ y,\ \frac{1}{2}\right]$	`[6,16]`
7	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[7,13]`
8	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[8,14]`

Table 9: Wyckoff site: `8i`, site symmetry: `.m.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ z\right]$	`[1,7]`
2	$\left[x+\frac{1}{2},\ \frac{3}{4},\ -z\right]$	`[2,8]`
3	$\left[-x,\ \frac{3}{4},\ -z\right]$	`[3,5]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ z\right]$	`[4,6]`
5	$\left[x,\ \frac{3}{4},\ z\right]$	`[9,15]`
6	$\left[x+\frac{1}{2},\ \frac{1}{4},\ -z\right]$	`[10,16]`
7	$\left[-x,\ \frac{1}{4},\ -z\right]$	`[11,13]`
8	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ z\right]$	`[12,14]`

Table 10: Wyckoff site: 8j, site symmetry: .m'.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ z\right]$	[1,15]
2	$\left[x+\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[2,16]
3	$\left[-x,\ 0,\ -z\right]$	[3,13]
4	$\left[\frac{1}{2}-x,\ 0,\ z\right]$	[4,14]
5	$\left[-x,\ \frac{1}{2},\ -z\right]$	[5,11]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z\right]$	[6,12]
7	$\left[x,\ 0,\ z\right]$	[7,9]
8	$\left[x+\frac{1}{2},\ 0,\ -z\right]$	[8,10]

Table 11: Wyckoff site: 8k, site symmetry: m..

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	[1,6]
2	$\left[\frac{3}{4},\ -y,\ -z\right]$	[2,5]
3	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	[3,8]
4	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	[4,7]
5	$\left[\frac{1}{4},\ y+\frac{1}{2},\ z\right]$	[9,14]
6	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ -z\right]$	[10,13]
7	$\left[\frac{3}{4},\ y,\ -z\right]$	[11,16]
8	$\left[\frac{1}{4},\ -y,\ z\right]$	[12,15]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$\left[x,\ y,\ z\right]$	[1]
2	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	[2]
3	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$\left[-x,\ -y,\ -z\right]$	[5]
6	$\left[\frac{1}{2}-x,\ y,\ z\right]$	[6]
7	$\left[x,\ \frac{1}{2}-y,\ z\right]$	[7]
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[8]
9	$\left[x,\ y+\frac{1}{2},\ z\right]$	[9]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[10]
11	$\left[-x,\ y,\ -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	[12]
13	$\left[-x,\ \frac{1}{2}-y,\ -z\right]$	[13]
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[14]
15	$\left[x,\ -y,\ z\right]$	[15]
16	$\left[x+\frac{1}{2},\ y,\ -z\right]$	[16]